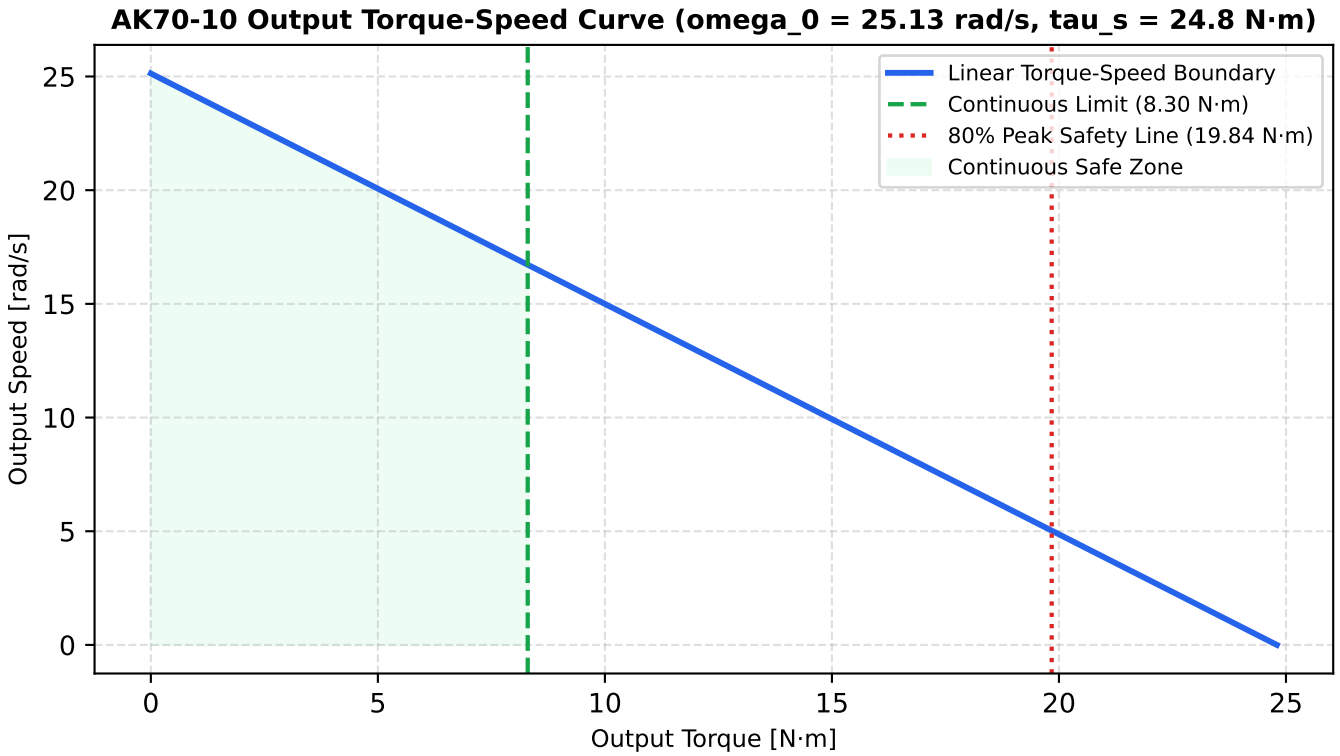
Part Number: AK70-10

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	AK70-10	AK-Series QDD
Manufacturer	Vendor	CubeMars	Outer-Rotor BLDC QDD
Rated Input Voltage	V_rated	24.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	24.8 N.m peak torque	At 24.0 V
Continuous Rated Torque	tau_cont	8.3 N.m rated torque	25 degC ambient, continuous duty
No-Load Speed	omega_0	240 rpm (at 24V KV100)	Unloaded output shaft
Actuator Mass (Weight)	m_act	521 g	Includes housing & internal gearhead
Internal Reduction	i_int	10 : 1 Planetary QDD	Integrated gear stage
Envelope Dimensions	W x H x D	80.0 x 50.0 x 80.0 mm	Outer housing envelope



2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 80.0 mm x 50.0 mm x 80.0 mm
   - Total Verified Mass: 521 g (0.5210 kg)
   - Mounting Pattern: M2.5 / M3 ISO radial & horn flange threads

3. Thermal & Duty-Cycle Notes
   - Continuous operation must remain at or below the continuous rated torque.
   - If continuous torque is not explicitly tabulated on Page 1, apply a 0.33x stall torque
   - Peak torque operation (>80% of available speed-dependent torque) risks step loss or the

4. Citation Verification Hash
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